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(54) ELECTRONIC MAIL TRANSFER AGENT WITH A PERSISTENT QUEUE, AND RELATED METHOD OF OPERATION

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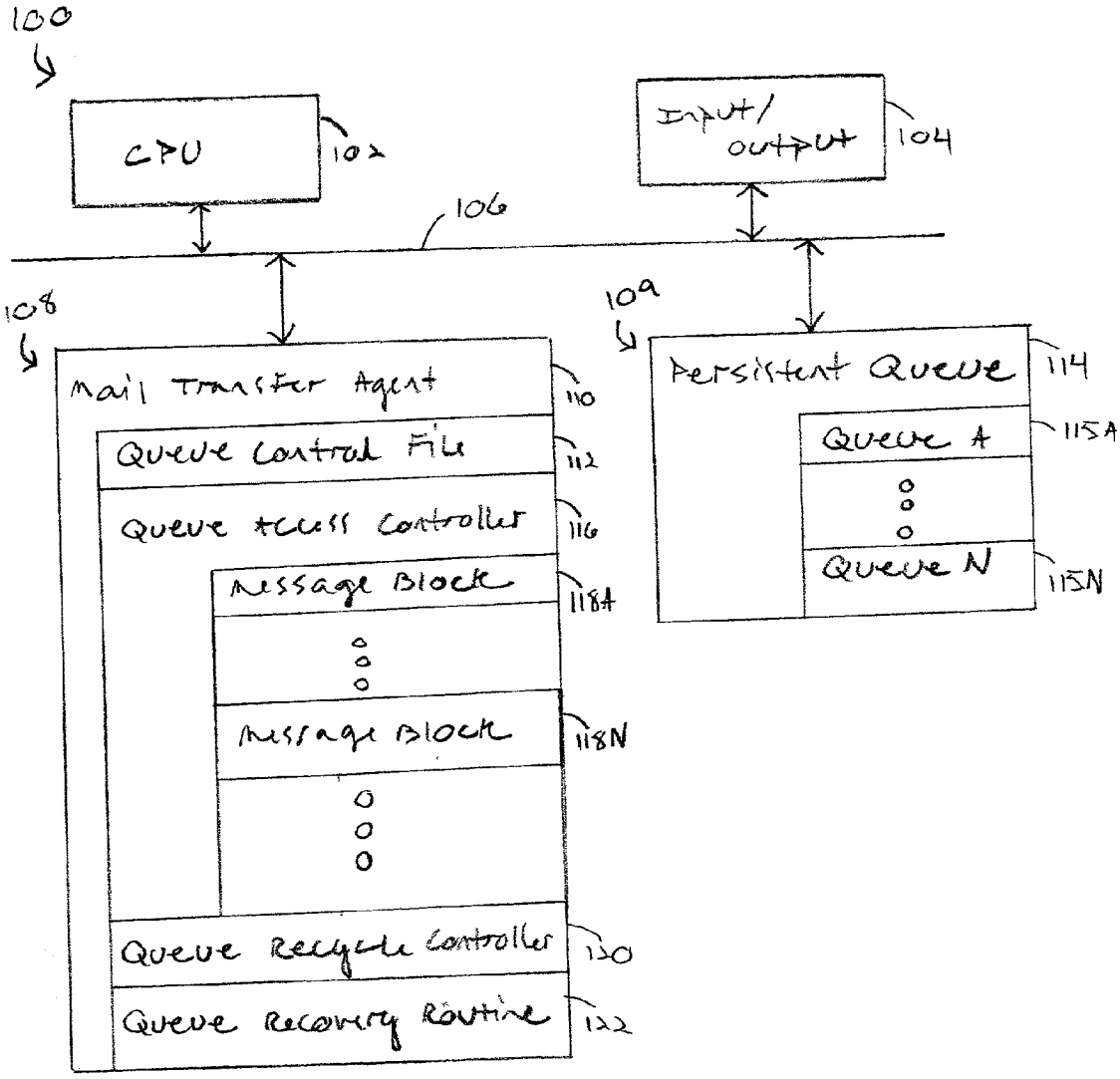
(57) ABSTRACT

A method of processing electronic mail messages includes configuring a queue on a permanent storage device. A set of electronic mail messages is accumulated to form a continuous electronic mail message block. The continuous electronic mail message block is stored in the queue in an uninterrupted write sequence.



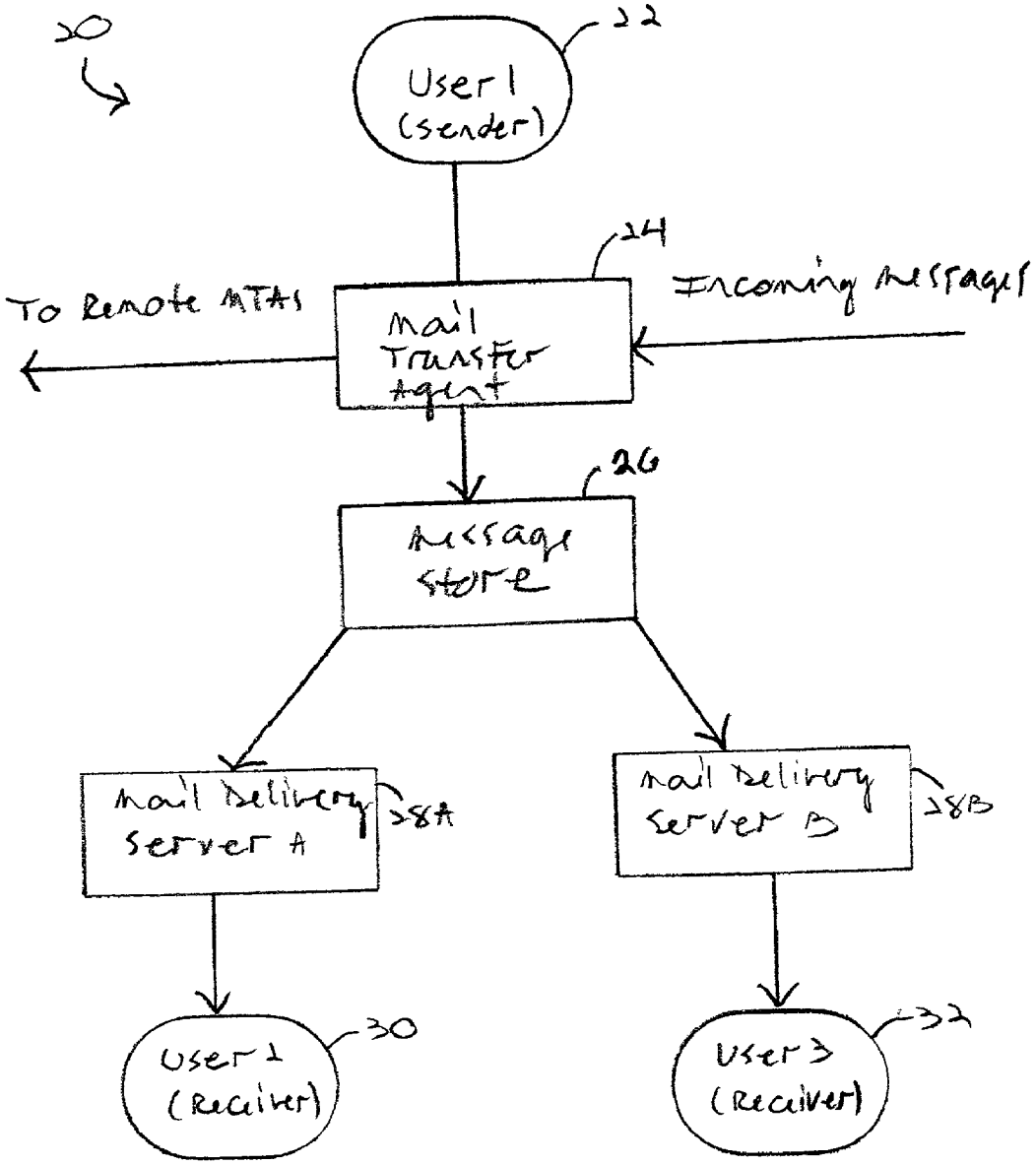



Fig. 1
(Prior Art)

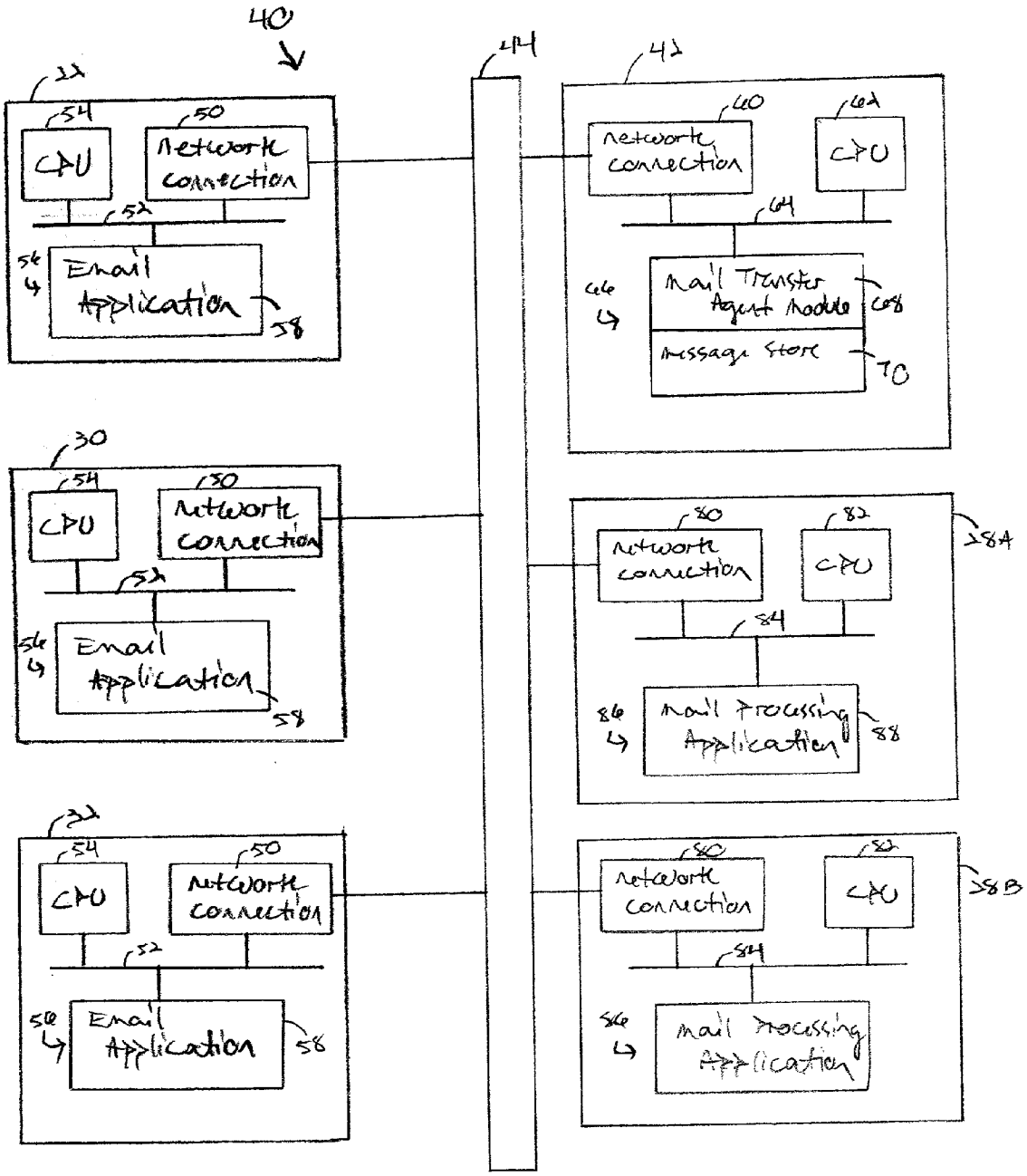



Fig. 2
(Prior Art)

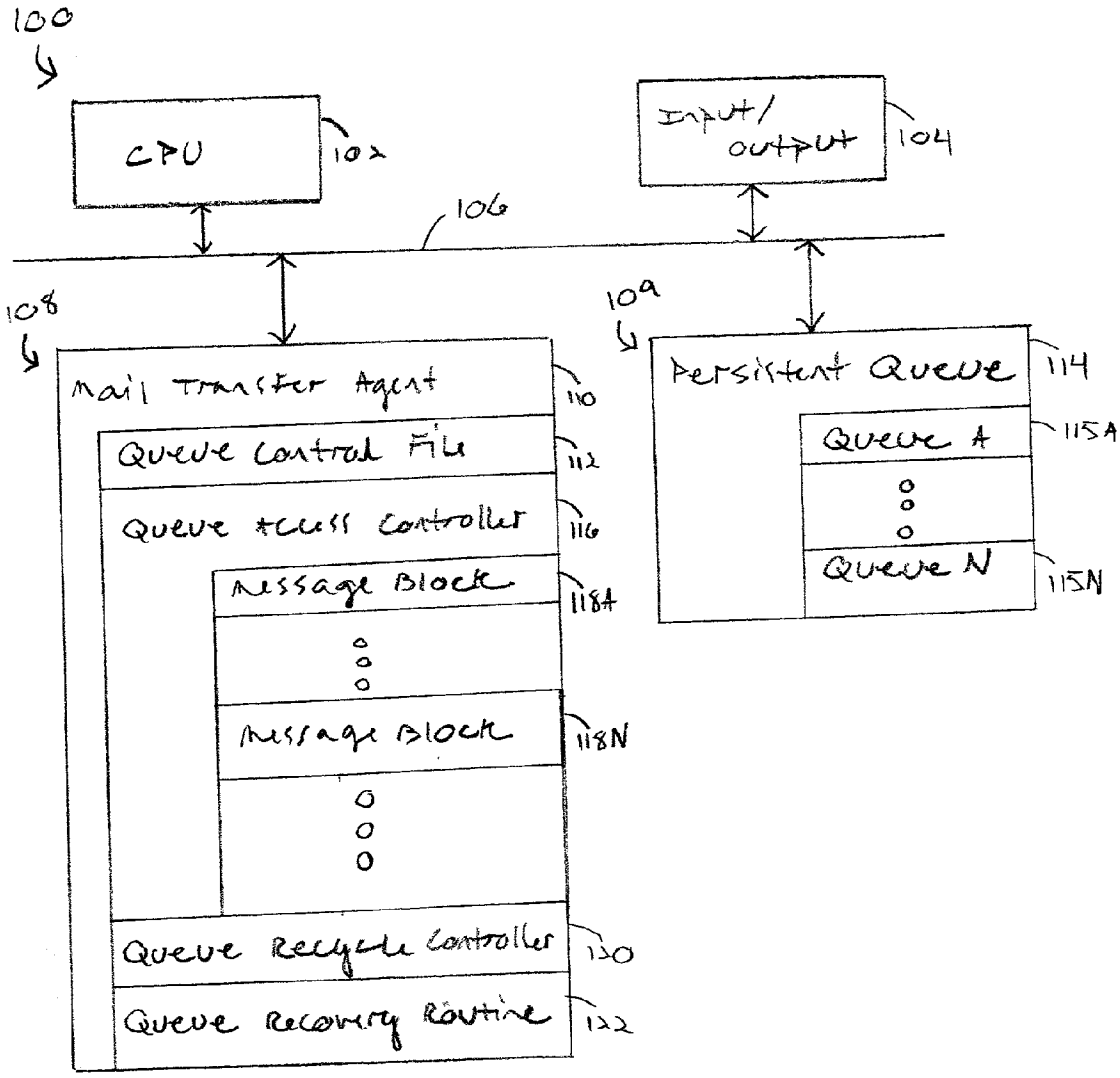

100
CPU
102
Input/
output
104
106
108
109
Mail Transfer Agent
110
Queue Control File
112
Queue Access Controller
116
Message Block
118A
Message Block
118N
Queue Recycle Controller
120
Queue Recovery Routine
122
Persistent Queue
114
Queue A
115A
Queue N
115N

Fig. 3

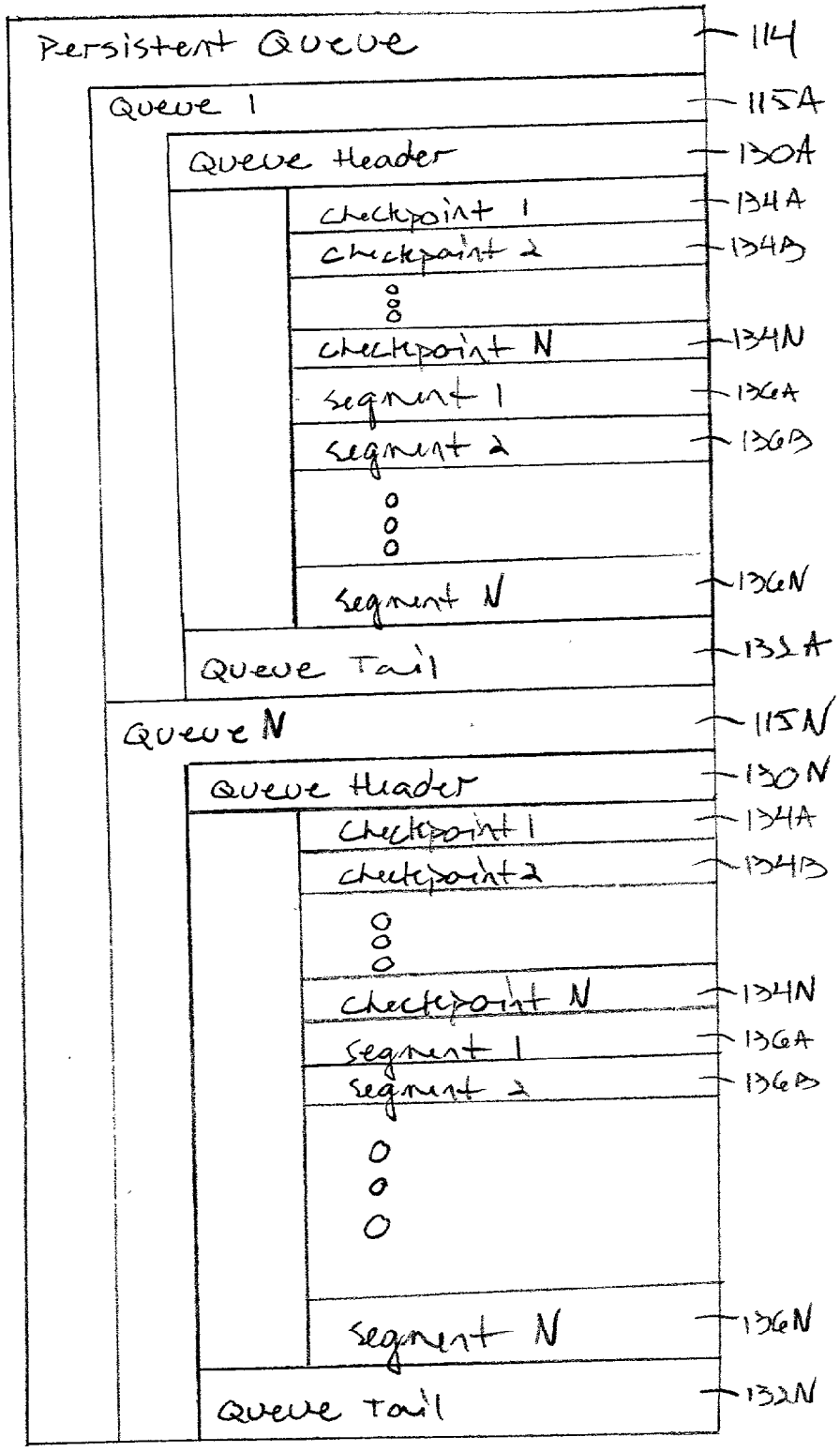

Persistent Queue
114
Queue 1
115A
Queue Header
130A
Checkpoint 1
134A
Checkpoint 2
134B
Checkpoint N
134N
Segment 1
136A
Segment 2
136B
Segment N
136N
Queue Tail
132A
Queue N
115N
Queue Header
130N
Checkpoint 1
134A
Checkpoint 2
134B
Checkpoint N
134N
Segment 1
136A
Segment 2
136B
Segment N
136N
Queue Tail
132N

Fig. 4

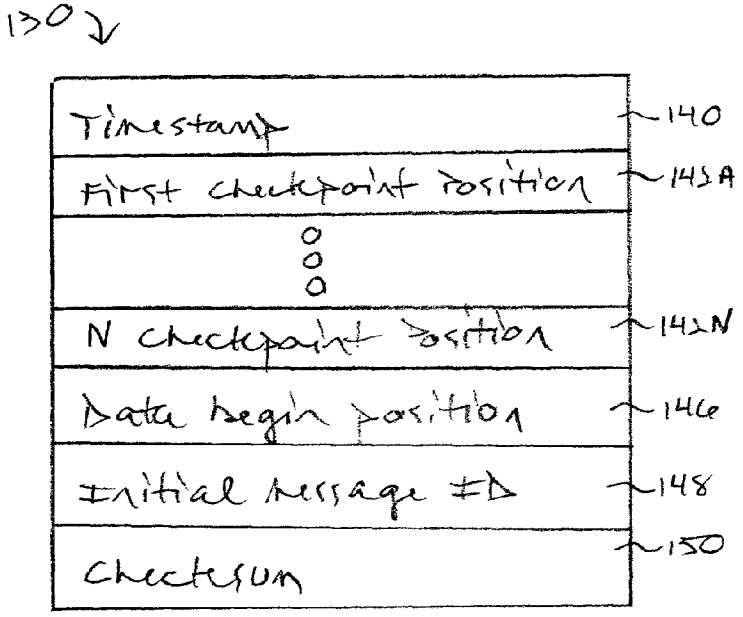

130
Timestamp
140
First checkpoint position
142A
N checkpoint position
142N
Data begin position
146
Initial message ID
148
Checksum
150

Fig. 5

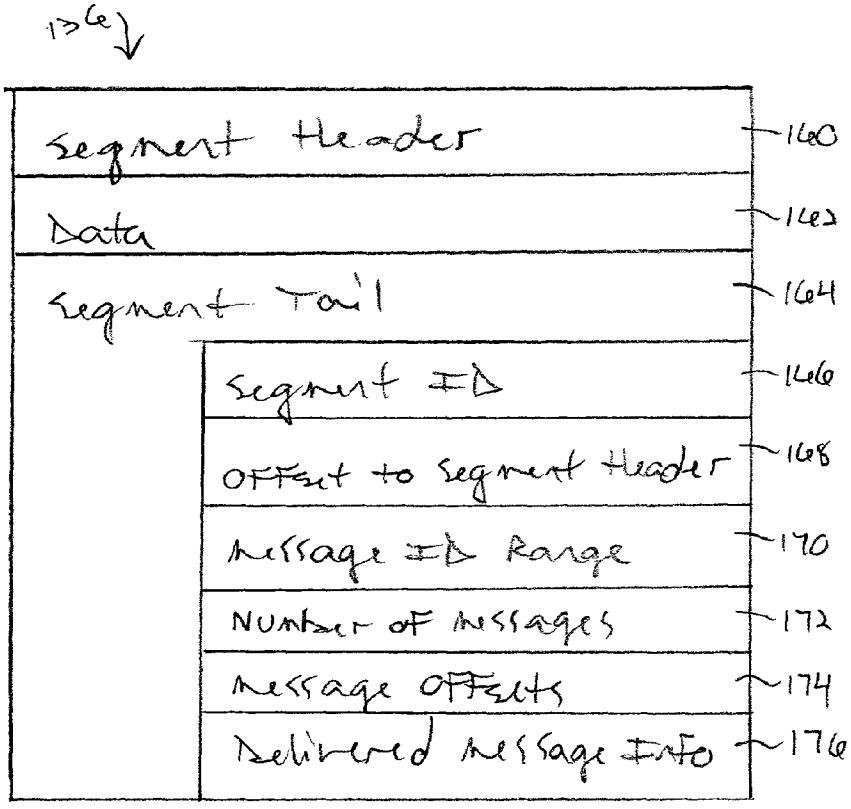

136
Segment Header
160
Data
162
Segment Tail
164
Segment ID
166
Offset to Segment Header
168
Message ID Range
170
Number of Messages
172
Message Offsets
174
Delivered Message Info
176

Fig. 6

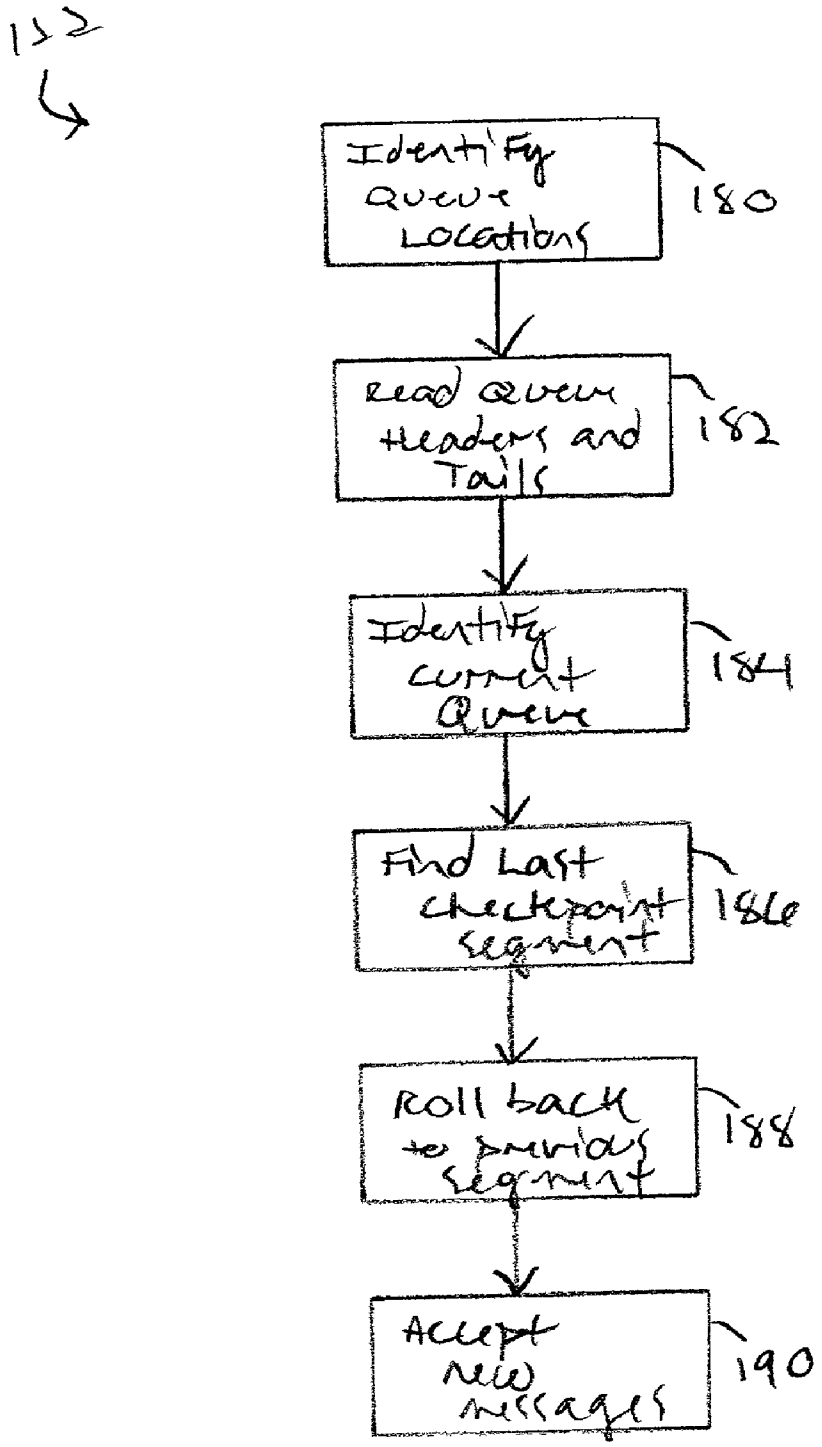

112
Identify Queue Locations
180
Read Queue Headers and Tails
182
Identify current Queue
184
Find Last checkpoint segment
186
Roll back to previous segment
188
Accept new messages
190

Fig. 7

# ELECTRONIC MAIL TRANSFER AGENT WITH A PERSISTENT QUEUE, AND RELATED METHOD OF OPERATION

## BRIEF DESCRIPTION OF THE INVENTION

[0001] This invention relates generally to the distribution of electronic mail in a networked environment. More particularly, this invention relates to a technique of utilizing a persistent queue in connection with an electronic mail transfer agent.

## BACKGROUND OF THE INVENTION

[0002] In a traditional postal system, a letter is sent to a first post office, which is responsible for sending the letter to another post office closer to the designated recipient. In an email system, the post office is called a Mail Transfer Agent (MTA). The protocol used between mail transfer agents is called the Simple Mail Transfer Protocol (SMTP).

[0003] **FIG. 1** illustrates a prior art electronic mail system **20**. A first user (e.g., User **1**) sends an electronic mail message from an electronic device **22**. The electronic mail message is processed by an MTA **24**. The MTA **24** has an associated message store **26** to persistently store the message. The first user at electronic device **22** cannot receive an acknowledgment from the MTA **24** until the MTA **24** writes the message into the message store **26**.

[0004] **FIG. 1** also illustrates a set of mail delivery servers **28**A and **28**B. The mail delivery servers may be standard servers, such as Post Office Protocol (POP) or Internet Message Access Protocol (IMAP) servers. A second user (e.g., User **2**) at an electronic device **30** accesses email messages through mail delivery server **28**A, while a third user (e.g., User **3**) at an electronic device **32** accesses email messages through mail delivery server **28**B. Once an electronic mail message is successfully delivered to a mail delivery server, it can be deleted from the message store **26**.

[0005] Thus, **FIG. 1** illustrates how a message generated by a first user at electronic device **22** is processed by an MTA **24** to facilitate the delivery of the message to a second user at electronic device **30** and a third user at electronic device **32**. As shown in **FIG. 1**, the MTA **24** processes other incoming messages and routes them to remote MTAs (not shown).

[0006] **FIG. 2** illustrates a computer network **40** corresponding to the system shown in **FIG. 1**. In particular, **FIG. 2** illustrates a first electronic device **22** connected to a server computer **42** through a network backbone **44**, which may be any wired or wireless data transmission infrastructure. Also connected to the network backbone **44** is a second electronic device **30** and a third electronic device **32**. As initially discussed in connection with the example of **FIG. 1**, the network **40** allows a first user at electronic device **22** to deliver a message to a second user at electronic device **30** and a third user at electronic device **32**. By way of example, each electronic device **22**, **30**, and **32** includes a network connection circuit **50** for interfacing with the network backbone **44**. The network connection circuit **50** is attached to a system bus **52**. Also attached to the system bus **52** is a central processing unit **54**. In addition, a memory **56** is connected to the system bus **52**. The memory **56** stores a standard email application program **58**.

[0007] The server **42** also includes a network connection circuit **60** and a central processing unit **62** connected via a system bus **64**. A memory **66** stores a mail transfer agent module **68**. The mail transfer agent module **68** includes a set of executable instructions to implement the functions of a mail transfer agent. The memory **66** also includes a message store **70** to store incoming electronic mail messages, as instructed by the mail transfer agent module **68**.

[0008] **FIG. 2** also illustrates a first mail deliver server **28**A and a second mail delivery server **28**B. Each mail delivery server includes a network connection circuit **80** connected to a central processing unit **82** via a system bus **84**. A memory **86** is also connected to the system bus **84**. The memory **86** stores a mail processing application **88**, for example implementing a POP or IMAP server.

[0009] Thus, as in the example of **FIG. 1**, if an email message is generated at the first electronic device **22**, it will be processed by the server **42**, which includes a mail transfer agent module **68** and a message store **70**. By way of example, the message may then be forwarded to server **28**A and server **28**B. After receiving acknowledgement from servers **28**A and **28**B, the copy of the message in the message store **70** may be deleted. The user at the second electronic device may then access the message from mail delivery server **28**A, while the user at the third electronic device may access the message from mail delivery server **28**B.

[0010] The various components discussed in connection with **FIGS. 1 and 2** are well known in the art. Unfortunately, there are a number or problems associated with these prior art systems. One significant problem associated with these prior art systems is that there is a bottleneck on synchronous disk input/output as incoming messages are queued. In prior art systems, each message is written to disk as a separate file. Once a message is stored in this manner, an acknowledgement can be sent to the previous computer that stored the message, allowing the previous computer to delete the message from its message store and/or notify the user that the message has been sent.

[0011] A large-scale email deployment requires the ability to handle thousands of incoming messages per second. These messages result in small synchronous random disk accesses, which cause spindle contention and input/output bottleneck. Current email systems try to alleviate this problem through different approaches. One approach is to deploy more mail transfer agents. This approach is expensive in terms of capital, management expense, and physical space requirements. Another approach is to implement a temporary message store on a network file system incorporated in a multi-disk system so that spindle contention is amortized across many disks. Many small writes across a network files system are inefficient. Therefore, even though this method may relieve spindle contention, overall performance is not increased. Still another approach is to implement a temporary message store on a local Redundant Array of Individual Disks (i.e., a RAID unit) so that spindle contention is amortized across many disks. Although this approach improves performance, servicing the RAID units on a set of mail transfer agents is difficult. In addition, RAID units for every mail transfer agent are prohibitively expensive.

[0012] In view of the foregoing, it would be highly desirable to provide an improved technique for storing electronic

mail messages in mail transfer agents. Such a technique could alleviate a substantial performance bottleneck that prevents current mail transfer agents from scaling to acceptably large sizes.

## SUMMARY OF THE INVENTION

[0013] The invention includes a method of processing electronic mail messages. A queue is configured on a permanent storage device or devices. A set of electronic mail messages or message portions are accumulated to form a continuous electronic mail message block. The continuous electronic mail message block is stored in the queue in an uninterrupted write sequence.

[0014] The invention also includes a computer readable memory to direct a computer to function in a specified manner. The computer readable memory includes a first set of instructions to configure a queue structure on a permanent storage device. A second set of instructions groups a set of sequentially received electronic mail messages into a continuous electronic mail message block. A third set of instructions produces a continuous disk write of the electronic mail message block to the queue structure.

[0015] The persistent queue of the invention provides fast input/output by aggregating small and scattered input/output events into one big, sequential input/output event. Sequential input/output can achieve raw disk input/output throughput, compared with small, scattered file system accesses, which can only obtain a small fraction of raw disk input/output throughput.

[0016] Another benefit associated with the invention is that it avoids fragmentation. After creating and deleting large numbers of files, the file system becomes fragmented. Fragmentation can slow down the overall system performance. The queue structure of the invention eliminates fragmentation because queues are re-cycled as a whole.

[0017] The checkpoint technique utilized in accordance with the persistent queue facilitates quick system recovery, allowing the mail transfer agent of the invention to operate in mission-critical applications. Advantageously, the invention can be implemented on relatively low cost hardware platforms, while still achieving high performance.

## BRIEF DESCRIPTION OF THE FIGURES

[0018] The invention is more fully appreciated in connection with the following detailed description taken in conjunction with the accompanying drawings, in which:

[0019] **FIG. 1** is a general illustration of a prior art electronic mail system.

[0020] **FIG. 2** illustrates a computer network implementing an electronic mail system.

[0021] **FIG. 3** illustrates a computer configured in accordance with an embodiment of the invention.

[0022] **FIG. 4** illustrates a persistent queue configured in accordance with an embodiment of the invention.

[0023] **FIG. 5** illustrates a queue header data structure that may be utilized in accordance with an embodiment of the invention.

[0024] **FIG. 6** illustrates a segment data structure that may be utilized in accordance with an embodiment of the invention.

[0025] **FIG. 7** illustrates a queue recovery routine that may be utilized in accordance with an embodiment of the invention.

[0026] Like reference numerals refer to corresponding parts throughout the several views of the drawings.

## DETAILED DESCRIPTION OF THE INVENTION

[0027] **FIG. 3** illustrates an electronic system **100** implemented in accordance with an embodiment of the invention. The electronic system **100** includes a central processing unit **102** connected to a set of input/output devices **104** via a system bus **106**. The input/output devices **104** may include a keyboard, mouse, video monitor, printer, network connection card, and the like. Also connected to the system bus are a primary memory **108** and a secondary memory **109**. The primary memory **108** stores a mail transfer agent **110**. The mail transfer agent **110** includes executable code to perform many of the functions performed by existing mail transfer agents. However, the mail transfer agent **110** also includes executable code to implement the operations of the present invention. In particular, the mail transfer agent **110** includes a queue control file **112**. The queue control file may include executable code and settings used to form a persistent queue **114** in the secondary memory **109** of the electronic system **100**.

[0028] When initially formed, the persistent queue **114** is a large empty file stored on one or more disks. The file can be a local file or a network file system file mounted on a storage server. As shown in **FIG. 3**, the persistent queue **114** may be implemented as a set of individual contiguous queues, such as queues **115**A-**115**N. Each queue has a fixed size and location, which is reserved after the queue is created. Preferably, the queue control file **112** specifies the number of queues and the location and size of each queue. The persistent queue **114** may be formed across a set of disks. In addition, the persistent queue **114** may be mirrored or replicated across a set of disks.

[0029] The memory **108** also stores a queue access controller **116**. The queue access controller **116** includes executable code that directs the mail transfer agent to accumulate a set of electronic mail messages to form a continuous electronic mail message block. Each message block **118**A-**118**N is preferably stored in primary memory and is then stored in the persistent queue **114** in an uninterrupted write sequence. By accumulating individual electronic mail messages into large groups and then writing the individual messages as a single set of information, disk accesses are reduced and high-speed raw disk writes can be achieved. A message block may contain portions of a single large message.

[0030] The memory **108** further stores a queue recycle controller **120**. The queue recycle controller **120** includes executable code to allow an individual queue (e.g., queue **115**A) to be over written after every message stored in the queue has been successfully processed. By preserving the queue space on the disk drive until this over write operation occurs, the persistent queue reduces disk fragmentation, which results in enhanced disk access speeds.

[0031] The memory **108** also stores a queue recovery routine **122**. The queue recovery routine **122** includes executable code to identify the checkpoint location corresponding to the last valid data segment stored in the persistent queue **114**. After a fault, this information is used for subsequent data writes to the persistent queue **114**, as discussed below.

[0032] **FIG. 4** is a more detailed illustration of a persistent queue **114** formed in accordance with an embodiment of the invention. The persistent queue **114** includes a set of individual contiguous queues **115**A-**115**N. Each queue **115** includes a queue header **130** and a queue tail **132**. Preferably, a set of checkpoints **134** are positioned between the queue header **130** and queue tail **132**. The checkpoints **134** are used to track the writing of information into the queue so that in the event of a system failure, the queue can be reconstructed. The queue also includes data segments **136**A-**136**N, which correspond to the electronic mail messages stored by the queue **114**. In one embodiment of the invention, each data segment **136** corresponds to one message block **118** from the queue access controller **116**.

[0033] **FIG. 5** illustrates a data structure that may be used to implement the queue header **130**. In one embodiment of the invention, the queue header **130** includes a timestamp field **140**, a first checkpoint position field **142**A, a position field for the Nth checkpoint **142**N, a data begin position field **146**, an initial message identification field **148**, and a checksum field **150**. The timestamp field **140** stores the time that the queue is formed. The first checkpoint field **142**A points to the position of the first checkpoint. Similarly, Nth checkpoint field **142**N points to the position of the Nth checkpoint. The data begin position field **146** points to the beginning of the data area in the queue. A data area includes a set of segments, which are described below. The initial message identification field **148** stores the identification of the initial message in the queue. The checksum field is used for validation.

[0034] The queue tail **132** is the last element in each queue. Preferably, the queue tail **132** includes a timestamp and a checksum. The timestamp and checksum for the queue tail will match for a valid queue.

[0035] As previously indicated, the checkpoints **134** are used to reduce recovery time. Each checkpoint records the last segment begin position. In one embodiment, two checkpoints are used in each queue. If the system crashes during the writing of one checkpoint, the other checkpoint can be used in recovery. Additional checkpoints can be used to reduce recovery time. However, writing checkpoints causes extra synchronous input/output, which reduces system throughput. The queue control file **112** can be used to specify the duration of each checkpoint.

[0036] **FIG. 6** illustrates a segment data structure that may be used in accordance with an embodiment of the invention. Data is stored in segments. Preferably, each write sequence to a disk causes one new segment to be created. If the current write sequence cannot fit into a single queue, the next queue is initialized and used. Preferably, each segment has three parts: a segment header **160**, data **162**, and a segment tail **164**. The segment header **160** points to the position of the segment tail **164**. The segment tail **164** preferably contains a segment identification field **166**, an offset to the segment header **168**, a message identification range **170**, a message number **172**, a message offset **174**, and delivered message information **176**. The delivered message information **176** specifies a low and high water mark. Every message below the low water mark has been delivered. Every message higher than the high water mark has not been delivered. A bitmap may be used to describe if the message in between has been delivered.

[0037] A large message may be fragmented into smaller pieces and then be written to the disk separately. The message fields **168**-**172** may be used to store information for this implementation.

[0038] Returning to **FIG. 3**, the memory **108** also stores a queue recovery routine **122**. **FIG. 7** illustrates processing steps that may be used to implement this operation. In the event of a system fault, the queue recovery routine **122** is invoked. The queue recovery routine **122** includes executable code to implement the following operations. First, the queue locations on disk are identified (**180**). This information may be obtained from the queue control file. The queue headers and queue tails are then read (**182**). The current queue is then identified using the queue header and/or queue tail timestamps (**184**). Once the current queue is identified, the checkpoint information in the queue is used to find the last checkpoint segment (**186**). The queue recovery routine **122** then rolls up to the previous segment in the queue (**188**). From this point forward, new messages may be accepted (**190**).

[0039] The mail transfer agent of the invention may be implemented in JAVA. Passing messages in a networked environment, manipulating strings, and processing database queries are strong implementation features associated with JAVA. A JAVA implementation also provides the benefits of extensibility, platform independence, and stability. Naturally, the invention may also be implemented using other programming languages. Alternately, the mail transfer agent of the invention may be implemented in a programmable logic device or as a hardwired circuit or as an application specific integrated circuit.

[0040] The persistent queue of the invention is general in nature. Therefore, its use can be extended to other implementations. For example, the techniques of the invention can be used to develop SMS or instant messaging applications on top of the message queue.

[0041] The foregoing description, for purposes of explanation, used specific nomenclature to provide a through understanding of the invention. However, it will be apparent to one skilled in the art that specific details are not required in order to practice the invention. Thus, the foregoing descriptions of specific embodiments of the invention are presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the invention to the precise forms disclosed; obviously, many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, the thereby enable other skilled in the art to best utilize the invention and various embodiments with various modifications as are suited to the particular use contemplated. It is intended that the scope of the invention be defined by the following claims and their equivalents.

In the claims:

**1**. A method of processing electronic mail messages, comprising:

configuring a queue on a permanent storage device;

accumulating a set of electronic mail messages to form a continuous electronic mail message block; and

storing said continuous electronic mail message block in said queue in an uninterrupted write sequence.

**2**. The method of claim 1 wherein configuring includes configuring said queue as a set of individual contiguous queues on said permanent storage device.

**3**. The method of claim 2 further comprising deleting a selected electronic mail message from said queue after said selected electronic mail message is successfully forwarded.

**4**. The method of claim 3 further comprising designating a selected queue of said set of individual contiguous queues for an overwrite operation when each electronic mail message in said selected queue has been successfully forwarded.

**5**. The method of claim 1 wherein configuring includes configuring said queue in accordance with a set of instructions specified in a queue control file.

**6**. The method of claim 1 wherein configuring includes configuring said queue to include a queue header and a queue tail.

**7**. The method of claim 6 wherein configuring includes configuring said queue to include a queue header specifying a time stamp field, a check point position field, a data segment begin position field, and a check sum field.

**8**. The method of claim 6 wherein configuring includes configuring said queue to include a set of checkpoints and data segments between said queue header and said queue tail.

**9**. The method of claim 8 wherein configuring includes configuring said queue to include data segments specifying a segment header field, a data field, and a segment tail field.

**10**. The method of claim 8 further comprising recovering from a fault by identifying a selected checkpoint location associated with the last valid data entry prior to said fault.

**11**. A computer readable memory to direct a computer to function in a specified manner, comprising:

a first set of instructions to configure a queue structure on a permanent storage device;

a second set of instructions to group a set of sequentially received electronic mail messages into a continuous electronic mail message block; and

a third set of instructions to produce a raw disk write of said continuous electronic mail message block to said queue structure.

**12**. The computer readable memory of claim 11 wherein said first set of instructions configures said queue as a set of individual contiguous queues on said permanent storage device.

**13**. The computer readable memory of claim 12 further comprising instructions to delete a selected electronic mail message from said queue after said selected electronic mail message is successfully forwarded.

**14**. The computer readable memory of claim 13 further comprising instructions to designate a selected queue of said set of individual contiguous queues for an overwrite operation when each electronic mail message in said selected queue has been successfully forwarded.

**15**. The computer readable memory of claim 11 wherein said first set of instructions includes instructions to configure said queue with a queue header and a queue tail.

**16**. The computer readable memory of claim 15 wherein said queue header includes a time stamp field, a check point position field, a data segment begin position field, and a check sum field.

**17**. The computer readable memory of claim 15 wherein said first set of instructions includes instructions to configure said queue with a set of checkpoints and data segments between said queue header and said queue tail.

**18**. The computer readable memory of claim 17 further comprising instructions to recover from a fault by identifying a selected checkpoint location associated with the last valid data entry prior to said fault.

* * * * *